FOOZHAN ATAIEFARD

Senior ML Researcher

+1(587)969-4222 | f.ataiefard@gmail.com | foozhan.github.io

SELECTED EXPERIENCES

Jul 2024 – Present **NVIDIA** ✧ *Senior Devtech Engineer/Deep Learning Research* – ***VLM, CV, Graphics***
- Generative Models, Diffusion-based rendering, Model optimization
- AI for games

Dec 2022 – Jul 2024 **Huawei Technologies Canada** ✧ *Senior AI Researcher* – ***LLM, CV, CUDA***
- Accelerate training performance and memory usage of deep learning models on GPU and NPU clusters using quantization, LoRA, various types of parallelism, and other model and communication compression techniques
- Develop and optimize CUDA kernels for custom PyTorch ops on AI accelerators

Models: LLama, GPT-3, ViT/DeiT, Swin, T5, Bert, Conformer
Frameworks: Deepspeed, Megatron-LM, Hugging face

Jul 2022 – Dec 2022 **Oracle Labs** ✧ *Student Researcher* – ***Reinforcement Learning, LLM***
- Fine-tuning LLMs using RLHF (PPO) for SQL code generation

Jul 2021 – Dec 2021 **Intelius AI** ✧ *Student Researcher* – ***DeepRL***
- Research on DNN Policies and reward functions robustness for A2C, PPO and DQN agents in stock trading
- Responsible for designing high-frequency trading RL agents

Jul 2020 – Oct 2022 **SEA-LAB** ✧ *University of Calgary, Research Assistant*
- My Research focused on adversarial policies to improve robustness of DeepRL trading agents

May 2019 – Oct 2020 **Itoll** ✧ *Machine Learning Scientist* – ***CV, OCR***
- Designed and Developed a verity of Computer Vision products including Live Object Tracking, Object Detection, Image Segmentation for one of country's largest ML startup
- Vehicle tracking for live CCTV cameras (real-time) using YOLO, and Faster-RCNN(Resnet50 backbone) regressor (following each track and velocity estimation)
and re-identifying occluded tracks using Siamese network with a triplet loss
- Hiring and leading a small team of Machine Learning researchers
- Deployment and scaling of Deep Learning products on IaaS Cloud servers
- Design and deployment of an OCR for vehicle license plate, currently reading $\sim$ 2 million license plates per day with accuracy of 96% (CRNN + CTC loss)

2016 – 2018 **AeLT Lab** ✧ *Amirkabir University of Technology*
Undergraduate Research Assistant
- Predicting user engagement using Information Retrieval methods in Social Media.
- Worked on hybrid recommender systems using Associated rule mining and Collaborative filtering
- Literature review on graph visualization methods
- Thesis: A survey on Graph Visualization

EDUCATION

2022 **University of Calgary**, Canada
M.Sc. in Software Engineering, GPA 4/4

2019 **Amirkabir University of Technology**, 1% acceptance rate
B.Sc. in Computer Science

2013 **Allameh Helli High School (NODET)**, 2% acceptance rate
Diploma in Mathematics and Physics Discipline

PUBLICATIONS

Single parent family: A spectrum of family members from a single pre-trained foundation model
H Hajimolahoseini, M Hassanpour, **F Ataiefard**, B Chen, Y Liu
under review

SkipViT: Speeding Up Vision Transformers with a Token-Level Skip Connection
F Ataiefard, W Ahmed, H Hajimolahoseini, S Asani, F Javadi, M Hassanpour, O Awad, A Wen, K Liu, Y Liu
Edge Intelligence Workshop, AAAI 2024

SwiftLearn: A Data-Efficient Training Method of Deep Learning Models using Importance Sampling
H Hajimolahoseini, O Awad, W Ahmed, A Wen, S Asani, M Hassanpour, F Javadi, M Ahmadi, **F Ataiefard**, K Liu, Y Liu
ENLSP workshop, Neurips 2023

GQKVA: Efficient Pre-training of Transformers by Grouping Queries, Keys, and Values
F Javadi, W Ahmed, H Hajimolahoseini, **F Ataiefard**, M Hassanpour, S Asani, A Wen, O Awad, K Liu, Y Liu
ENLSP workshop, Neurips 2023

Gray-box Adversarial Attack of Deep Reinforcement Learning-based Trading Agents
F Ataiefard, H Hemmati
ICMLA 2023

Deep State Inference: Toward Behavioral Model Inference of Black-box Software Systems
F Ataiefard, MJ Mashhadi, H Hemmati, N Walkinshaw
IEEE TSE 2021

A Hybrid Approach for Thread Recommendation in Massive Open Online Course Forums
A Kardan, A Narimani, **F Ataiefard**
IJCSE 2017

SKILLS

Frameworks	Pytorch, CUDA, OpenAI Gym, Stable Baselines, RLlib, TensorFlow, Keras, Numpy, Scikit-Learn
Programming Languages	Python, C/C++, SQL
OTHER	Linux, Vim, Redis, git, CI/CD, Competitive programming

HONORS & AWARDS

2022 **NSERC CREATE Research Scholarship**, University of Calgary

2016 **4th place in ACCEPT2 ACM-ICPC coding contest**

2011 **1st place in State, 4th place National in Robotics**, 13th Khwarizmi International Award

2011 **4th place in RoboCup IranOpen**
Junior Soccer Cup**(Qualified for RoboCup 2011)**